

Module 6 Task 1: AI Adoption Initiative Design Canvas

Initiative: Blankinship Lab AI Onboarding Program (soil science research lab, University of Arizona). Inputs reviewed per instructions: Module 3 Workflow Canvas and Module 5 Unit-Level AI Acceptable Use Guideline.

SECTION 1 - PROBLEM STATEMENT (50 words max)

Each semester the Blankinship Lab onboards new graduate and undergraduate researchers who already use AI for code, literature, and drafting, but no one trains them on our new acceptable-use guideline. Untrained use risks unverified facts in analyses, sensitive-data exposure, and undisclosed AI in publications across a 7-member lab.

SECTION 2 - TARGET AUDIENCE

Role(s): incoming and current lab members: graduate students, undergraduates, postdocs, lab technician.

AI fluency (1-5): mixed, mostly 2-3 (daily chatbot users with little verification discipline); one or two 4s.

What they need: one-hour, hands-on rules for tools, data classes, verification, and disclosure they can apply the same day.

Key concern they have: that new rules will slow down or restrict the AI use they already rely on.

SECTION 3 - AI TOOLS AND APPROACH				
Tool	Modality	Rationale	Data sensitivity	Approved?
Claude (desktop app)	Text/multimodal chat	Drafting and debugging analysis code, editing own text; the assistant members already use.	Low-sensitivity inputs only	No
Perplexity (Gemini 3.1 Thinking)	AI search with inline citations	The lab's documented background-research workflow (Module 3 canvas); citations speed verification.	Low only	No
Lab citation-verification tool (in-house)	Local software checker	Automates the guideline's verify-before-use rule for cited literature in manuscripts.	Low (published papers)	N/A (lab-built, runs locally)

SECTION 4 - ETHICAL SAFEGUARDS		
Risk	Safeguard mechanism	Responsible party
Unpublished or sensitive data entered into unapproved tools	Data-sensitivity classification taught in the module; signed guideline acknowledgment; posted approved-tool list.	Lab manager
Hallucinated facts or citations entering research	Verify-before-use rule (every fact traced to its original source); citation checker run on all manuscripts.	Every member; lab manager spot-audits
Undisclosed AI use in publications	AI-disclosure line on the manuscript submission checklist; AI-use log archived with each data package.	PI (Dr. Blankinship) at submission
AI quality-control scripts silently excluding data points	Flags stay recommendations; a human signs off on every exclusion with a logged reason code.	Lab manager

SECTION 5 - EQUITY CONSIDERATIONS

Population 1: undergraduate and hourly assistants without paid AI subscriptions or capable personal laptops. **Adjustment:** all exercises run on free tiers and the shared lab workstation; nothing requires a personal purchase.

Population 2: international students and other non-native English speakers. **Adjustment:** plain-language materials, a recorded version, and worked examples of the permitted “AI as editor of your own text” use.

Also monitored: field-based members with limited connectivity get the recorded, asynchronous version.

SECTION 6 - SUCCESS METRICS

Metric	How measured	Timeline	Minimum acceptable threshold
Onboarding completion + signature	Roster vs. signed acknowledgment forms	Within 2 weeks of joining; reported each semester	100%
Verification competence	5-scenario quiz at the end of the module	Every onboarding session	Average $\geq 4/5$; anyone below 3/5 retakes
Disclosure compliance	Submission-checklist audit of every manuscript and thesis	At each submission; reviewed Day 90, then semesterly	100% carry a disclosure plus archived AI-use log
Perceived usefulness	3-question anonymous pulse after each session	Each semester	$\geq 80\%$ rate it useful, otherwise revise

SECTION 7 - ROLLOUT: FIRST THREE STEPS

Step 1: draft the one-hour module (slides, 5-scenario quiz, signature form) from the Module 5 guideline. **Who:** me (lab manager). **Resource:** existing guideline; Claude for drafting. **By:** 17 July 2026 (this week; no approvals needed).

Step 2: review with Dr. Blankinship at lab meeting, revise, and add the signature form to the lab onboarding SOP. **Who:** me + Dr. Blankinship. **Resource:** 30-minute lab-meeting slot. **By:** 31 July 2026.

Step 3: pilot with the incoming fall cohort and all current members; collect quiz scores and pulse feedback. **Who:** me. **Resource:** 1-hour lab-meeting slot; printed forms. **By:** 4 September 2026.

30-SECOND ELEVATOR PITCH (50 words max, written last)

Our lab already wrote its AI ground rules; this program makes them stick. A one-hour onboarding module each semester teaches every new researcher which tools and data are allowed, how to verify AI output, and how to disclose it, so our science stays trustworthy while members work faster. (48 words)